

FitBuddy Requirement Analysis

Comprehensive Functional, Non-Functional, Hardware & Software Technical Specifications

01. Requirement Analysis Overview

The requirement analysis phase serves as a vital step in the engineering process of the **FitBuddy** platform. Its main objective is to identify and analyze user needs to establish concrete feature requirements and operational bounds. FitBuddy is designed as a streamlined fitness companion to manage workouts, set goals, track activities, and view continuous progress in one accessible platform.

Major Identified Requirements:

User Requirements

Intuitive interface design enabling simple, friction-free access to all core fitness utilities.

Workout Management

Ability to view, filter, and select structured exercises according to personal fitness goals.

Fitness Goal Setting

Capabilities to set custom targets and systematically monitor completion progress.

Activity Tracking

Simple recording mechanisms for logging daily completed workouts and active effort.

Progress Monitoring

Clear historical visualization of workout logs, achievements, and consistency metrics.

Central Dashboard

Unified hub granting rapid navigation to workout plans, target metrics, and active records.

Health & Fitness Guidance

Curated tips for general physical activity, rest, hydration, and healthy lifestyle habits.

Data & Interface Management

Proper data storage paired with a fully responsive layout supporting multi-device access.

02. Types of Requirements

A. Functional Requirements

Functional requirements specify the direct capabilities and computational workflows provided by the FitBuddy application.

Requirement	Functional Specification
User Registration & Login	Allows users to create accounts and securely authenticate to access personalized fitness data.
Workout Plans	Delivers comprehensive exercise routines, sets, reps, and workout guides.
Goal Management	Provides controls to set, edit, track, and complete individual target metrics.

Requirement	Functional Specification
Activity Recording	Enables real-time and post-workout logging of physical activities and exercises.
Progress Tracking	Generates visual output and chronological logs of historic workout data.
Central Dashboard	Consolidates active goals, workout summaries, and quick navigation links into one primary screen.
Fitness Guidance	Displays rotating actionable tips for healthy living, hydration, and recovery.
Profile Management	Allows users to modify personal attributes, account settings, and fitness preferences.

B. Non-Functional Requirements

Non-functional requirements specify quality benchmarks, technical constraints, and usability operational criteria.

Quality Attribute	Specification Standard
Usability	Simple, clean visual design minimizing cognitive load and interface complexity.
Performance	Fast page load times and fluid responsive UI interactions without perceptible delay.
Reliability	Consistent system behavior, uninterrupted features, and reliable data retention.
Security	Protection of sensitive personal parameters against unauthorized access.
Compatibility	Full cross-browser compatibility (Chrome, Edge, Firefox) and adaptive screen responsiveness.
Maintainability	Modular codebase enabling simple code maintenance, updates, and debugging.
Accessibility	Readable font typography, clear contrast ratios, and structured layout navigation.
Scalability	Extensible architecture built to accommodate future feature expansions seamlessly.

03. Technical Specifications (Hardware & Software)

Hardware Requirements	Software Requirements
Processor: Dual-core processor or higher	Operating System: Windows, macOS, or Linux
RAM: 4 GB minimum (8 GB recommended)	Core Tech: HTML5, CSS3, JavaScript (ES6+)
Storage: Adequate storage for project files & assets	IDE / Editor: Visual Studio Code
Devices: Desktop, Laptop, Tablet, Smartphone	Web Browsers: Google Chrome, Edge, Firefox
Connectivity: Active Internet connection for web features	Version Control: Git & GitHub

04. Expected Outcome

The requirement analysis phase delivers a complete operational blueprint covering functional capabilities, quality attributes, and technical environments for FitBuddy. This specification serves as the formal baseline for system architecture design, UI/UX prototyping, and software implementation, ensuring FitBuddy delivers a reliable, accessible, and high-performance user experience.